

Notes: Maksimovic and Titman (RFS, 1991)

Financial Policy and Reputation for Product Quality

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1 Big picture

Maksimovic and Titman (1991) develop a theory in which financial policy affects product-market behavior because it changes a firm's incentive to preserve its reputation for product quality. The paper is not mainly about direct bankruptcy costs. It is about an indirect cost of financial distress: customers may expect a financially weak or highly levered firm to cut hidden product quality, so they reduce the price they are willing to pay. This expected loss of reputation lowers firm value even before formal bankruptcy occurs.

The product-quality friction is simple but powerful. Customers cannot verify quality before purchase and cannot enforce quality in court. A firm can reduce quality to save current costs, but doing so reveals that it is a flexible type willing to cheat and damages future sales. An all-equity firm may maintain quality because future reputation rents are valuable. Debt changes the payoff to equityholders: it makes current cash flows more valuable relative to future cash flows because some future cash flows accrue to debtholders or disappear in bankruptcy. Therefore, leverage can make quality cutting privately attractive even when it destroys total firm value.

A central insight is that debt is not always bad for reputation. When assets have a valuable liquidation option, all-equity shareholders may be tempted to liquidate rather than preserve a costly reputation. Properly chosen senior claims can commit equityholders not to liquidate in some states, or can shift liquidation control to debtholders in states where liquidation is efficient. Thus, depending on asset redeployability and financial condition, leverage can either destroy or support reputation.

2 Incremental contribution of the paper

1. **Endogenizes indirect bankruptcy costs through product-market reputation.** The paper explains why customers, suppliers, or other stakeholders may avoid a financially weak firm even if they do not directly lose from bankruptcy.
2. **Links capital structure to hidden product quality.** Debt changes the shareholders' incentive to honor implicit contracts. Quality choice becomes a financing-policy problem.
3. **Shows why dividend covenants are incomplete.** Dividend restrictions can limit some asset-substitution incentives, but if quality cuts raise reported current earnings, standard dividend covenants may not prevent hidden quality reductions.
4. **Separates financial shortfall from financial distress.** A firm with a shortfall can borrow and preserve quality if its condition is known and it can raise funds. A firm in true financial distress may cut quality to avoid immediate bankruptcy.
5. **Introduces an "involuntary customer loan" interpretation.** Cutting hidden quality raises current cash flows by borrowing from customers through a lower future reputation value. This implicit claim is senior to existing debt, so it dilutes debtholders.
6. **Shows leverage can increase reputation credibility when assets are redeployable.** Debt can reduce inefficient liquidation incentives and can implement a first-best liquidation/quality policy in some states.
7. **Broadens capital-structure theory beyond taxes and direct distress costs.** Optimal leverage depends on product-market implicit contracts, asset specificity, liquidation values, and

the observability of financial condition.

3 Main research question

The paper asks:

How does debt financing affect a firm's ability to credibly maintain a reputation for high product quality when quality is imperfectly observable and cannot be contractually enforced?

The question matters because many markets rely on implicit contracts. Airlines must maintain safety, food producers must maintain purity, hospitals and drug companies must maintain quality, and suppliers must trust firms to behave fairly. The paper argues that these product-market reputations interact with capital structure. A firm that looks financially weak may be distrusted by customers not because bankruptcy itself harms them, but because financial weakness changes the firm's incentives before bankruptcy.

4 Model ingredients

4.1 Quality and firm types

The firm sells one unit in each of three periods. Product quality can be high or low. High quality costs c , while low quality is normalized to have zero production cost. Customers value high quality more, but they cannot verify quality before purchasing and cannot enforce high quality in court.

There are two types of firms:

- **Type H:** constrained or committed to produce high quality.
- **Type F:** flexible; can choose high or low quality and maximizes shareholder value.

Customers do not observe the type directly. If a firm has never been caught producing low quality, customers assign probability λ_t that the firm is type F in period t . If a firm produces low quality, its type is revealed and reputation is lost.

4.2 Prices and reputation

The price of a product in period t is

$$p_t = \theta_t - \delta_t l,$$

where θ_t is the price of a product known to be high quality, l is the loss in value from low quality, and δ_t is the customer's assessed probability that the product is low quality. A higher perceived probability of low quality lowers the price immediately.

This pricing rule makes reputation valuable. Producing high quality today can raise tomorrow's price by keeping customers' posterior belief favorable. Producing low quality saves c today but reduces future prices.

4.3 Why finite horizon matters

If customers knew with certainty that the firm is a flexible shareholder-value-maximizing type, backward induction would eliminate reputation in a finite horizon. The firm would cut quality in the last period, customers would anticipate that, reputation would have no value in the penultimate period, and the unraveling would continue to the first period.

The two-type structure prevents this full unraveling. Since customers do not know whether a firm is type H or type F, a type F firm may mimic the type H firm by producing high quality to preserve the belief that it may be committed.

5 The all-equity benchmark

In the all-equity model, the firm has no debt and no chance of bankruptcy. The type F firm compares the current cost saving from low quality to the future price loss from losing reputation.

The key second-period comparison is:

$$\begin{aligned} V_2^h(\theta_2) &= \theta_2 - c + E_2[\theta_3] - \lambda_3 l, \\ V_2^l(\theta_2) &= \theta_2 + E_2[\theta_3] - l. \end{aligned}$$

Producing high quality in period 2 is attractive if the cost c is less than the reputation benefit $(1 - \lambda_3)l$. A lower posterior probability that the firm is flexible makes reputation more valuable because customers attach a higher probability to the firm being committed.

5.1 Interpretation of Proposition 1

Proposition 1 divides the all-equity equilibrium into three cases based on the prior probability λ_0 that the firm is flexible.

If λ_0 is low, reputation is easy to sustain. Customers believe the firm is likely to be type H, so a type F firm has strong incentive to mimic type H by producing high quality in the first two periods. It cuts quality only in the final period, where reputation has no continuation value.

If λ_0 is intermediate, the firm produces high quality in the first period but mixes in the second. The intermediate case captures a point where reputation is valuable enough to sustain early quality but not strong enough to ensure quality in the penultimate period.

If λ_0 is high, the firm mixes already in the first period. Customers suspect the firm is flexible, so mimicking is less rewarding. Still, the equilibrium cannot have all type F firms cut quality immediately, because then producing high quality would become a strong positive signal. Mixed strategies make the firm exactly indifferent.

6 Debt without liquidation: reputation can be weakened

6.1 Debt obligations and equity payoffs

The levered firm has debt obligations d_2 and d_3 due at the end of periods 2 and 3. Equity has limited liability. New financing, if raised, is assumed to be junior to existing debt. This seniority

assumption is central: it means that issuing new debt can be costly to equityholders because existing debtholders receive priority.

A firm is bankrupt if it cannot meet debt obligations even after cutting quality. The paper distinguishes two weaker but important conditions:

- **Financial shortfall:** the firm needs external financing to produce high quality, but can obtain financing.
- **Financial distress:** the firm cannot raise enough external capital to finance high quality and can avoid immediate bankruptcy only by cutting quality.

6.2 Future bankruptcy and dividend restrictions: Proposition 2

Proposition 2 revises the earlier claim that future bankruptcy alone makes firms cut quality. If the firm can immediately pay all current earnings as dividends, then high future debt can make low quality attractive: equityholders capture the current cost saving while future losses fall partly on debtholders.

However, if dividends and debt payments in period 2 cannot exceed the cash flow generated by high-quality production, the firm cannot transfer the quality-cutting gain to shareholders immediately. Under that restriction, future debt alone does not change the quality choice relative to all equity. The interpretation is close to Myers's debt-overhang model: payout policy determines whether equityholders can capture the private benefit of underinvestment or quality cutting.

The important qualification is that standard dividend covenants may not solve the product-quality problem. If reducing hidden quality increases reported earnings, then an earnings-based dividend covenant may actually allow larger payouts after quality is cut.

6.3 Financial shortfall with known condition: Proposition 3

When financial condition is observable and the firm has a shortfall that can be covered by borrowing, Proposition 3 says the firm will borrow rather than cut quality. The reason is informational. If customers can observe that the firm did not borrow despite needing funds for high quality, they infer that it will produce low quality. The price then immediately falls, eliminating the short-run benefit of the hidden quality cut.

This result is subtle: debt does not automatically reduce quality when customers can observe the financing problem and the firm can raise enough funds. The firm protects its reputation by borrowing, and the act of borrowing credibly communicates high-quality intentions.

6.4 Financial distress: Proposition 4

Proposition 4 studies a firm that cannot finance high-quality production. If the quality cost saving is large enough relative to the market's posterior belief, the firm can avoid immediate bankruptcy by cutting quality. The condition

$$c > (\lambda_2 - \delta_2)l$$

compares the private cost saving from low quality to the price/reputation penalty embedded in customer beliefs.

The mechanism is an indirect bankruptcy cost. A firm in distress may cut safety, service, maintenance, or other hidden quality dimensions to survive. Customers rationally anticipate this incentive, which can reduce the price paid to the firm and deepen distress.

6.5 Unobservable financial condition: Proposition 5

If customers do not observe whether the firm has debt obligations, a financially weak but not distressed firm may cut quality rather than borrow. Borrowing on junior terms can be unattractive because new lenders demand compensation for risk and existing senior debt captures much of the upside from preserving the firm's future cash flows.

Cutting quality becomes a substitute source of financing. It is like borrowing from customers: the firm obtains current resources by delivering lower quality than customers expected, and the repayment is the future reputation loss. This implicit customer claim is effectively senior to existing debt, so it transfers value away from debtholders toward equityholders.

6.6 Period-1 quality choice: Proposition 6

The reputation problem can unravel backward through the possibility of future shortfalls. Even if there is no chance of financial distress or bankruptcy in period 2, a levered firm may have a stronger incentive than an all-equity firm to cut quality in period 1. The reason is that the firm anticipates it may need costly junior financing later. Reducing quality in period 1 supplies cash earlier and avoids future financing costs.

This proposition is important because it shows that the quality cost of leverage is not confined to firms already near default. Leverage can reduce reputation incentives today because it changes the expected cost of maintaining quality in future states.

6.7 Senior versus junior debt

The paper assumes new financing is junior to existing debt. If the firm could issue senior secured debt, some quality-cutting incentives would be mitigated because the firm could raise funds by diluting existing bondholders more efficiently than by cutting quality. This parallels Stulz and Johnson's secured-debt logic for underinvestment.

But the fix is temporary. Once new senior debt is in place, the firm can again have incentives to cut quality to expropriate the new creditors. The deeper problem is the conflict between current equityholders and all future claimants, including customers.

7 Liquidation option: debt can improve credibility

7.1 The liquidation problem in an all-equity firm: Proposition 7

Section 3 adds an asset liquidation option. The firm can liquidate in period 2 after the state is revealed but before production. This option seems valuable because it gives the firm flexibility. But it can reduce reputation value. If an all-equity firm can liquidate, customers know that the

firm may exit rather than keep investing in reputation. This weakens the incentive to produce high quality in period 1.

Proposition 7 states that the liquidation option weakly reduces the probability of high-quality production in period 1, and strictly reduces it in the high- λ_0 mixed-strategy region. The value of the firm with the option can be lower than the value without it. This formalizes the Klein and Leffler intuition that firms may commit to low salvage value to make reputation credible.

7.2 Debt as commitment and first-best liquidation: Proposition 8

Debt can solve part of the liquidation problem. If senior claimants receive liquidation proceeds before equityholders, then equityholders may be less tempted to liquidate merely to capture salvage value. In simple cases, issuing debt whose face value exceeds liquidation value commits equityholders not to liquidate.

Proposition 8 goes further. Under specific conditions, debt can implement a policy that mimics the all-equity firm with zero liquidation value for product-quality incentives while still allowing liquidation in states where liquidation is efficient. The idea is that the firm becomes bankrupt precisely in states where liquidation is optimal, so debtholders liquidate when liquidation creates value, while equityholders preserve reputation incentives in operating states.

This is the main positive role for leverage in the paper. Debt is not only a source of agency costs; it can also allocate control rights in a way that strengthens the credibility of implicit contracts.

8 Economic interpretation and predictions

8.1 When debt is especially costly

Debt is most damaging when quality is hidden, reputation is important, customers observe or infer financial weakness, and quality cutting raises current cash flows. Examples include airlines, food and pharmaceuticals, hospitals, warranty-intensive goods, suppliers with service obligations, and any firm that depends on customers trusting implicit promises.

The theory predicts that a firm in these industries may choose lower leverage, stronger payout restrictions, more conservative liquidity, or governance/compensation contracts that reduce short-term cash-flow manipulation.

8.2 When debt can be valuable

Debt can be useful when asset redeployability is high and the main reputation problem is that equityholders might liquidate too easily. In those settings, debt can commit equityholders to continuation or can transfer control to senior claimants in states where liquidation is socially efficient. This makes the effect of debt depend on whether assets are firm-specific or have strong alternative uses.

8.3 Empirical implications

The model suggests several empirical patterns:

- Firms with high reputation sensitivity should suffer larger product-market losses after financial distress.
- Quality outcomes should deteriorate when firms become financially weak, especially when quality is difficult to verify immediately.
- Customers should demand discounts from highly levered firms in markets with hidden quality.
- Dividend covenants tied only to reported earnings may be weak when firms can temporarily boost earnings by cutting hidden quality.
- The effect of leverage should vary with asset redeployability: debt may be less damaging, or even beneficial, when collateral has high salvage value.

9 Relation to later literature

This paper anticipates several later research themes. It is an early contribution to product-market theories of capital structure, along with Titman (1984), Brander and Lewis (1986), and Maksimovic's work on oligopoly and leverage. It also foreshadows stakeholder theories of financial distress in which suppliers, employees, and customers respond to balance-sheet weakness. Finally, the idea that collateral, liquidation values, and implicit contracts interact with capital structure connects naturally to later work on asset specificity, collateral constraints, fire sales, and financial contagion.

10 Limitations and assumptions

The paper is theoretical and does not provide a direct empirical test. The model uses a finite three-period horizon, rational and informed customers, and a two-type firm structure. It assumes that quality is observable after purchase but not verifiable in court, and it often assumes debt cannot be renegotiated. These assumptions are useful for clarity, but real markets may contain additional mechanisms such as warranties, regulation, repeat intermediaries, reputation platforms, renegotiation, and managerial compensation contracts.

The paper also abstracts from the possibility that some debt covenants, warranties, inspections, or certification systems could directly prevent quality reductions. Its most robust lesson is not that debt always reduces quality, but that financial policy changes the credibility of implicit product-market promises.

11 What to remember

The key takeaway is that capital structure affects more than investment and payout. It also affects whether customers believe the firm will honor noncontractible quality promises. Leverage can shift value from future reputation and debtholders to current equityholders, making hidden quality

cutting tempting. But leverage can also serve as a commitment device when liquidation options undermine reputation. The sign of the effect depends on the interaction between debt seniority, payout restrictions, financial distress, information about firm condition, and asset liquidation value.

12 Panels and interpretation

12.1 Panel 1: Title, Abstract, and Central Claim; Panel 2: All-Equity Quality Choice and Equations (1)-(2)

Financial Policy and Reputation for Product Quality

Vojislav Maksimovic
University of British Columbia

Sheridan Titman
University of California, Los Angeles

The effect of financial policy on a firm's incentives to maintain its reputation for producing a high-quality product is analyzed. It is demonstrated that in certain situations debt will reduce a firm's ability to credibly offer high-quality products and, as a consequence, will reduce its value. However, for firms with assets that have high salvage values in liquidation, debt may increase their ability to credibly offer high-quality products and, therefore, increase their values.

Panel 1: Title, abstract, and central claim.

future. Since reputation has no value after the last period, a type r firm will always find it advantageous to produce low-quality goods in period 3.

Suppose that in period 2 the customers believe that the firm will produce high quality, so that if it does, its profits will be $\theta_2 - c$, while if it produces low quality its profits will be θ_2 . The firm will choose to produce high quality in period 2 if the expected sum of its profits in periods 2 and 3 from producing high quality in the current period, $V_b^2(\theta_2)$, exceeds the expected profits from producing low quality, $V_l^2(\theta_2)$, where

$$V_b^2(\theta_2) = \theta_2 - c + E_2[\theta_3] - \lambda_3 l, \quad (1)$$

$$V_l^2(\theta_2) = \theta_2 + E_2[\theta_3] - l, \quad (2)$$

and $E_{t-1}[\theta_t]$ is the expectation of θ_t .

Panel 2: All-equity quality choice and equations (1)-(2).

Interpretation. Panel 1 states the paper's central contribution in compressed form: debt may reduce the ability to credibly offer high quality, but debt can also help when assets have high salvage values. Panel 2 sets up the all-equity reputation tradeoff. Producing high quality costs c today, while producing low quality saves the cost but triggers a future price/reputation penalty. The comparison between equations (1) and (2) is the basic reputation condition that all later leverage results modify.

12.2 Panel 3: Proposition 1 - All-Equity Reputation Equilibrium; Panel 4: Proposition 2 - Future Bankruptcy, Dividends, and Quality Choice

Proposition 1. (i) If $\lambda_0 < 1 - (c/l)$, there exists a unique sequential equilibrium in which a type F firm produces high quality in the first two periods and low quality in the last period.

(ii) If $1 - (c/l) < \lambda_0 < 1 - (c/l)^2$, there exists a unique sequential equilibrium in which a type F firm always produces high quality in the first period and produces high quality in the second period with probability π_2 , where $\pi_2 = (1 - \lambda_0)(l - c)/c\lambda_0$. The firm always produces low quality in the last period.

(iii) If $\lambda_0 > 1 - (c/l)^2$, there exists a unique sequential equilibrium in which a type F firm produces high quality in the first period with probability $\pi_1 = (1 - \lambda_0)(l^2 - c^2)/\lambda_0 c^2$. If it produced high quality in the first period, the firm produces high quality in the second period with probability $\pi_2 = c/(l + c)$; otherwise, it produces low quality. The firm always produces low quality in the last period.

Panel 3: Proposition 1 - all-equity reputation equilibrium.

Proposition 2. For d_3 sufficiently large (but not large enough that the firm is sure to go bankrupt), a type F firm will produce a low-quality product in period 2 if it pays out all of its period 2 earnings in the form of a dividend. If in period 2 the firm can pay no more than its profit from producing a high-quality product, $p_2 - c$, in debt payments and dividends, it will produce high quality in this period with the same probability as an all-equity firm, regardless of its period 3 debt obligation.

Proof. We prove this for the case in which the all-equity firm produces a high-quality product in the first and the second periods with probability 1. As shown in Proposition 1, this occurs if $\lambda_0 < 1 - (c/l)$. The proof for the case in which $\lambda_0 > 1 - (c/l)$ is similar, and is omitted in the interest of brevity.

The following beliefs are shown to be consistent if $\lambda_0 < 1 - (c/l)$: the type F firm will produce a high-quality product in periods 1 and 2 and a low-quality product in period 3. Given this belief, if the firm produces high quality, its equity value, $S_2^H(\theta_2)$, is

$$S_2^H(\theta_2) = \theta_2 - c - d_2 + E_2[\max(\theta_3 - \lambda_0 l - d_3, 0)]. \quad (3)$$

If the firm cannot pay more than $\theta_2 - c - d_2$ in dividends, then its equity value from producing low quality is

$$S_2^L(\theta_2) = \theta_2 - c - d_2 + E_2[\max(\theta_3 - l + c - d_3, 0)]. \quad (4)$$

A comparison of the above equilibrium values reveals that $S_2^H(\theta_2) \geq S_2^L(\theta_2)$ as long as $\lambda_0 < 1 - (c/l)$, where the inequality is strong as long as there is some probability that the firm producing high quality does not go bankrupt. Hence, the assumed beliefs are consistent.

If the firm can pay its entire income as a dividend,

$$S_2^H(\theta_2) = \theta_2 - d_2 + E_2[\max(\theta_3 - l - d_3, 0)], \quad (5)$$

Panel 4: Proposition 2 - future bankruptcy, dividends, and quality choice.

Interpretation. Proposition 1 is the clean all-equity benchmark. The lower the prior probability that the firm is flexible, the more valuable reputation becomes and the more likely high quality is sustained. Proposition 2 shows that high future debt by itself is not enough to force low quality if payout is restricted. The damaging incentive appears when equityholders can capture the current quality-cutting savings through dividends while future losses fall partly on debtholders.

12.3 Panel 5: Proposition 3 - Financial Shortfall with Known Financial Condition; Panel 6: Proposition 4 - Financial Distress and Quality Cutting

Proposition 3. *If its financial condition is known, a type F firm with a financial shortfall that can be covered with additional borrowing will not reduce quality.*

Proof. We prove this for the case in which the all-equity firm produces a high-quality product in the first and the second periods with probability 1. As shown in Proposition 1, this occurs if $\lambda_0 < 1 - (c/l)$. The proof for the case in which $\lambda_0 > 1 - (c/l)$ is similar and is omitted in the interest of brevity.

Let the customers believe that the firm will produce high-quality output in period 2 if it borrows and not otherwise. If the amount of the shortfall, $s \equiv d_2 - (\theta_2 - c)$, is such that

$$s < (1 - \lambda_0)E_2[\max(\theta_3 - \lambda_0 l - c - d_3, 0)] + \lambda_0 E_2[\max(\theta_3 - \lambda_0 l - d_3, 0)], \quad (6)$$

the firm can finance its period 2 production by borrowing. We first
Panel 5: Proposition 3 - financial shortfall with known financial condition.

be unable to borrow or, alternatively, its financial condition must be unknown. Now consider the case in which the firm's financial condition is known, but because its current shortfall producing high quality $(\theta_2 - c - d_2)$ exceeds the present value of its period 3 net cash flows, it is in financial distress and faces certain bankruptcy unless it reduces the quality of its product. In this case, there can exist no equilibrium in which the type F firm produces high quality. The financially distressed type F firm will produce low quality and in some cases will still go bankrupt, while in other cases it will avoid bankruptcy. The following proposition provides conditions under which it will in fact be able to avoid bankruptcy in financial distress.

Proposition 4. *A firm that is in financial distress can avoid immediate bankruptcy by reducing quality if and only if $c > (\lambda_2 - \delta_2)l$, where δ_2 is the probability that an all-equity type F firm that has produced high quality in the first period produces low quality in the second period, and*

$$\delta_2 = \begin{cases} 0, & \text{if } \lambda_0 < 1 - c/l, \\ \frac{\pi_1(1 - \pi_2)\lambda_0}{1 - \pi_1 + \pi_2\lambda_0}, & \text{if } \lambda_0 > 1 - c/l. \end{cases}$$

Panel 6: Proposition 4 - financial distress and quality cutting.

Interpretation. Proposition 3 separates a temporary financing shortfall from true distress. If the firm can borrow and customers know its condition, not borrowing is a bad signal and destroys the immediate benefit of cutting quality. Proposition 4 studies the harsher case: if the firm cannot finance high quality, it may avoid bankruptcy by cutting quality. This is the core product-market distress channel.

12.4 Panel 7: Proposition 5 - Unobservable Financial Condition; Panel 8: Proposition 6 - First-Period Quality Choice with Future Financing Shortfalls

expects. As the following proposition demonstrates, a firm may, in this case, choose to reduce quality rather than borrow, because the cost associated with borrowing at unfavorable terms more than offsets the benefits of credibly revealing its intention to produce high quality by raising additional funds.

Proposition 5. Assume the following.

(i) With probability $\kappa > 0$, the firm has debt obligations d_2^* and d_3^* such that the firm is not in financial distress but that risky borrowing is required to finance high-quality production in the second period. With probability $1 - \kappa$ the firm has no debt obligations.

(ii) Customers do not observe whether or not the firm has debt obligations.

Then, if the second period price is sufficiently low, there is a positive probability that a type F firm will reduce quality rather than borrow to finance high-quality production.

Proof. To prove the proposition, assume the converse, that the firm produces high quality with certainty. Then the firm will receive θ_2 for its product whether or not it borrows. Let $s = d_2^* - (\theta_2 - c)$ be the shortfall. To borrow this amount, the firm must take on an additional obligation, d_n , which is junior to d_3^* , where d_n solves the equality

$$s = (1 - \lambda_0)E_2[\min(d_n, \max(\theta_3 - \lambda_0 l - c - d_3^*, 0))] + \lambda_0 E_2[\min(d_n, \max(\theta_3 - \lambda_0 l - d_3^*, 0))]. \quad (9)$$

The terminal value of a type F firm's equity if it produces low quality in the second period is

$$S_l^2 = E_2[\max(\theta_2 - d_2^* + \theta_3 - l - d_3^*, 0)]. \quad (10)$$

Panel 7: Proposition 5 - unobservable financial condition.

firm produces low quality, the sum of these period 1 and 2 cash flows is

$$p_1 + \theta_2 - l \quad (12)$$

If the firm produces high quality in period 1 and gets into financial distress, and produces low quality in period 2, the sum of its period 1 and 2 cash flows is

$$p_1 - c + \theta_2 - \lambda_2 l \quad (13)$$

Comparing expressions (12) and (13), we see that if $\lambda_2 > 1 - (c/l)$, and if the probability of financial distress is sufficiently high, the firm is better off producing low quality initially. However, the motivation to reduce quality can be much stronger than this if there is some probability that the firm may have to raise capital in the future. As we show in the next proposition, the firm may have an incentive to reduce quality in period 1 even if there is no chance that it will suffer financial distress in period 2.

Proposition 6. A levered type F firm may have a greater tendency to reduce quality in period 1 than an all-equity firm, even if there is no possibility of financial distress or bankruptcy in period 2.

Proof. This proposition is proved by presenting an example in which a levered firm that does not have sufficient debt obligations to be in danger of financial distress produces low quality with a higher equilibrium probability in the first period than an all-equity firm. The example assumes (i) $\lambda_0 > 1 - (c/l)^2$, so that an all-equity type F firm produces high-quality output with a probability less than 1 in periods 1 and 2; (ii) a two-state density in the final period where $\theta_3 = \theta_1 + a$ or $\theta_3 = \theta_1 - a$ and where the two states are equally likely; (iii) that the state of the world in the second period is known with certainty (i.e., $\theta_2 = \theta_1$); (iv) that d_2 is such that a firm that has produced low quality in the first period has a zero cash flow net of debt repayment in the first two periods (i.e., $d_2 = p_1 + \theta_1 - l$); (v) that d_n is sufficiently high that the type F firm becomes bankrupt if the price declines in the third period regardless of its quality decision; and (vi) that $l > 2c$. Given these assumptions, the value of the equity of a firm that produces low quality in period 1 is given by

$$S_l^1 = \frac{1}{2}[p_1 + \theta_1 - l + \theta_1 + a - l - d_2 - d_n]. \quad (14)$$

Panel 8: Proposition 6 - first-period quality choice with future financing shortfalls.

Interpretation. Proposition 5 shows how asymmetric information about financial condition can make quality cutting attractive even outside immediate distress. Borrowing on junior terms may be too costly for equity, so the firm secretly cuts quality instead. Proposition 6 pushes the logic backward: the prospect of future shortfalls can reduce quality today. This is why reputation losses can arise before default becomes imminent.

12.5 Panel 9: Proposition 7 - Liquidation Option and Reputation; Panel 10: Proposition 8 - Debt, Liquidation, and First-Best Commitment

3. The Option to Liquidate

In the preceding analysis the firm's assets had no alternative use and, hence, had a liquidation value of zero. If the firm's assets do have an alternative use, it may choose to liquidate its assets rather than continue to produce. In the following analysis we assume that the firm has the opportunity to liquidate its assets in the second period, before production takes place but after the state of nature has been revealed. Although increasing the opportunities available to a firm generally increases its value, in the present context the option to liquidate can have the adverse effect of decreasing the firm's ability to offer a high-quality product. This is shown in the following proposition.

Proposition 7. *If an all-equity firm has the opportunity to liquidate in period 2, it will (weakly) reduce its probability of producing high quality in the first period. When $\lambda_0 > 1 - (c/l)^2$, it will always reduce the probability of producing high-quality goods. The value of a firm that has the opportunity to liquidate may be lower than that of a firm that does not.*

Panel 9: Proposition 7 - liquidation option and reputation.

states of nature where it would optimally want to liquidate. This will not necessarily lead to the first-best outcome, in which the firm produces at the quality level of an all-equity firm without the liquidation option, since the debt level that allows the firm to be bankrupt in period 2 states of nature in which liquidation is optimal may also create a potential for financial distress in other states in which immediate bankruptcy can be avoided by reducing quality. However, as the following proposition demonstrates, under certain conditions this first-best solution can be achieved.

Proposition 8. *Assume that the following conditions are satisfied:*

- (i) $E_2[\theta_3 | \theta_2]$ is increasing in θ_2 ,
- (ii) $d_2 = p_1 - c + \theta^H - \delta_2 l - c$ and $d_3 = E_2[\theta_3 | \theta^H] - \delta_3 l$,
- (iii) $c < (\lambda_2 - \delta_2)l$,

where δ_2 and δ_3 are the customers' assessments of the probability that the good is of low quality for a firm that has an all-equity financial structure and does not have a liquidation option, and where θ^H is the period 2 state of nature in which an all-equity type H firm is indifferent between operating and liquidating. Then there exists an equilibrium where the firm will produce high quality with the same probability as an all-equity firm with a zero liquidation value, and will liquidate when doing so is optimal.

Proof. The proposition is proved by showing that the customers' belief, that the type F firm described in the proposition produces high-quality output with the same probability as an all-equity firm without the liquidation option, is consistent with the firm's strategies given the belief. Specifically, we show that δ'_2 and δ'_3 , the probabilities that the

Panel 10: Proposition 8 - debt, liquidation, and first-best commitment.

Interpretation. Proposition 7 is the paper's bridge from negative debt effects to positive debt effects. A liquidation option can make an all-equity firm less credible because it may exit instead of protecting reputation. Proposition 8 shows how debt can restore commitment by assigning liquidation payoffs and control rights differently across states. This is the main reason the paper does not simply conclude that leverage is always harmful.

12.6 Panel 11: Proposition 8 Proof and Equations (16)-(17); Panel 12: Conclusion on Financing, Reputation, and Product Markets

it follows from Proposition 1 that the sum of the period 2 and 3 cash flows for a type H firm and for a type F firm, which has produced low quality in period 1, are equal. As a result, they will liquidate in the same states. A type F firm that has produced high quality in the first period will also liquidate in exactly these states in order not to reveal its type. Thus, since $E_1[\theta_3 | \theta_2]$ is increasing in θ_2 , the firm, regardless of type, maximizes value by always liquidating when θ_2 is below θ^H .

If d_2 just exceeds $p_1 - c + \theta^H - \delta'_2 l - c$ and $d_3 = E_1[\theta_3 | \theta_2 = \theta^H] - \delta'_3 l$, then the type F firm cannot finance production of the high-quality good from internal funds for $\theta_2 \leq \theta^H$ and cannot borrow to do so either. If $c < (\lambda_2 - \delta'_2)l$, then the firm also cannot remain solvent by cutting quality when it is unable to borrow. Thus, the firm will become bankrupt in the second period for $\theta_2 < \theta^H$ and will be optimally liquidated by the debtholders.

The period 1 value of the firm's equity, conditional on it producing high quality in the first period, is given by

$$S^H_1 = qE_1[E_2[\max(p_1 - c + \theta_2 - \delta'_2 l + \theta_3 - l - d_2 - d_3, 0) | \theta_2 > \theta^H], \quad (16)$$

where $q = \Pr(\theta_2 > \theta^H)$. The equity value achieved by producing low quality is given by

$$S^L_1 = qE_1[E_2[\max(p_1 + \theta_2 - l + \theta_3 - l - d_2 - d_3, 0) | \theta_2 > \theta^H], \quad (17)$$

If the firm is to produce high quality in period 1 with a probability between 0 and 1, then $S^H_1 = S^L_1$. Equating expressions (16) and (17) yields $c + \delta'_2 l = l$. Similarly, the equilibrium condition $S^H_2(\theta_2) = S^L_2(\theta_2)$ for $\theta_2 > \theta^H$ yields $c + \delta'_3 l = l$. Since the restrictions on δ'_2 and δ'_3 are identical to the equilibrium restrictions on δ_2 and δ_3 for the all-equity firm without the liquidation option derived in Proposition 1, it follows that the beliefs that $\delta'_2 = \delta_2$ and $\delta'_3 = \delta_3$ are consistent with the equilibrium. ■

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The above proposition indicates that debt need not have a negative affect on a firm's ability to credibly offer high-quality products. In industries in which capital goods have high salvage values, debt may serve to commit the firm to produce higher quality goods than it might otherwise produce.

i. Conclusion

In this article, a theory has been proposed in which the effect of debt financing on a firm's ability to maintain a reputation for product quality

Panel 12: Conclusion on financing, reputation, and product markets.

Panel 11: Proposition 8 proof and equations (16)-(17).

Interpretation. Panel 11 shows the formal logic behind the first-best debt result: the relevant high- and low-quality equity values are compared only in states where the firm continues rather than liquidates. Panel 12 states the paper's broad message: product-market reputation is a determinant of capital structure, and the same logic applies beyond product quality to suppliers, employees, competition, and other implicit contracts.

13 Compact proposition map

Result	Main claim	Economic interpretation
Proposition 1	All-equity reputation depends on the prior probability that the firm is flexible.	Reputation is strongest when customers believe the firm may be committed to high quality.
Proposition 2	Future debt induces low quality only when shareholders can extract current gains through dividends.	Payout restrictions matter, but standard earnings-based restrictions may be incomplete.
Proposition 3	If financial condition is known and borrowing is feasible, the firm borrows and preserves quality.	Not borrowing would reveal low quality and destroy the benefit of cheating.
Proposition 4	A distressed firm can avoid immediate bankruptcy by cutting quality if the cost saving is large enough.	Financial distress can cause hidden quality deterioration.
Proposition 5	If financial condition is unobservable, the firm may cut quality rather than borrow.	Hidden quality cuts are an implicit loan from customers and dilute debtholders.
Proposition 6	Leverage can reduce quality in period 1 even without future bankruptcy.	Anticipated future shortfalls can undermine present reputation.
Proposition 7	A liquidation option can reduce first-period quality and firm value.	Flexibility can weaken reputation commitment.
Proposition 8	Debt can implement high quality and efficient liquidation under conditions.	Senior claims can improve commitment when assets have high salvage value.